

Xiang Zhao

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RESEARCH SUMMARY

Research focus: controllable visual generation, with an emphasis on precise editing under preservation constraints.

Current interests: structured perceptual priors, VL/VLM-based supervision, and multimodal reward design for fine-grained visual editing and video generation.

Long-term direction: exploring how multimodal video generation can connect to action-conditioned modeling and VLA-style systems.

EDUCATION

Northeastern University

M.Eng. | Control Engineering

Shenyang, China

2016.09 – 2019.01

Courses: System Engineering & Decision Analysis, Applied Mathematics Statistics, Optimal Method & Theory, Computer Vision, Control System Design

Northeastern University

B.Eng. | Automation

Shenyang, China

2012.08 – 2016.06

Courses: Linear Algebra, Probability Theory & Mathematical Statistics, Machine Learning & Intelligent Optimization, Microcomputer Principle & Program Design

PROFESSIONAL EXPERIENCE

Alibaba Hujing Digital Media & Entertainment Group

Senior Algorithm Engineer — Video & Image Generation

Beijing, China

2023.08 – Present

Real-Time Lip Sync for Ultra-Realistic Digital Humans

- **Architecture:** Built end-to-end data pipeline and inpainting lip-generation model on a StyleSync backbone; introduced contrastive loss between generated lip shapes and speech features to improve lip-speech alignment.
- **Personalization:** Designed automated per-identity fine-tuning requiring only ~1 min of reference video.
- **Efficiency:** Post-distillation inference on a single RTX 4090 reduced from 1 s to 0.5 s per video-second; deployed in production for digital-cabin and customer-service scenarios.

Expression Editing via Reinforcement Learning Fine-tuning

- **Training pipeline:** Pre-trained on paired rendered data for supervised alignment, then fine-tuned on real face data via Flow-GRPO RL to bridge the synthetic-to-real gap.
- **VLM-guided reward:** Designed a mixed reward combining VLM-based emotion alignment with face-ID similarity, enabling fine-grained expression control without pixel-level supervision on real images.
- **Model merging & outcome:** Blended RL and SFT checkpoints via parameter interpolation to balance identity preservation with expression removal; improved on both dimensions over the internal baseline.

Audio-Driven Lip Sync on a Video Diffusion Foundation Model

- **Localized control:** Introduced Gaussian-weighted spatial classifier-free guidance centered on the mouth region during teacher inference, enabling localized lip editing without explicit masks; the student network internalizes this spatial bias through distribution matching distillation.
- **Training strategy:** Used a multi-stage curriculum with high-noise steps for structural reconstruction and low-noise steps for fine-grained lip detail synthesis; additionally explored VL-assisted annotation and richer video-side supervision for improved temporal and semantic control.
- **Research motivation:** This line of work sparked an interest in whether prediction-based video representations could strengthen temporal coherence and provide a path toward action-conditioned modeling.
- **Efficiency:** Reduced diffusion inference from 50 to 4 steps via Distribution Matching Distillation (DMD); domain-adapted model showed strong performance on children's animation relative to the baseline.

Microsoft Xiaoice

Senior Algorithm Engineer — Computer Vision

Beijing, China

2022.06 – 2023.07

High-Fidelity Face Swapping for Virtual Idols

- **Identity Compensation Network:** Designed an auxiliary control pathway to recover structural identity information discarded by standard face-recognition embeddings, improving fine-grained identity preservation.
- **FaceBlender:** Developed a StyleGAN-based blending network via self-supervised learning to reduce color discrepancy after face fusion; deployed for virtual idol content generation on Douyin.

BIGO Technology

Senior Algorithm Engineer

Beijing, China

2020.05 – 2021.08

Unpaired Style Transfer for Real-Time Mobile Deployment

- **Distillation & inference:** Trained image-to-image style transfer models (Disney, CG face) using an inverse StyleGAN approach to construct pseudo-paired training data without real pairs; compressed via knowledge distillation for real-time mobile execution.

- **Deployment:** Deployed StyleGAN custom operators to Triton, reducing per-frame latency from 90 ms to 80 ms; models packaged with stickers without requiring app updates.

Multi-Person Drivable Photo Singing Sticker

- **System:** Led training of 512-resolution driving model and mobile-distilled inference network as project owner; achieved ~3 s for 116-frame single-person driving on device.
- **Impact:** First sticker supporting simultaneous multi-person driving; reached 100k+ DAU in the Southeast Asian market during the launch period.

TAL Education Group

Algorithm Engineer

Beijing, China
2019.01 – 2020.03

Lip Sync for Recorded Lectures via 3D Parametric Face Model

- **Method:** Mapped voice to 20-dim 3D expression parameters; trained GAN-based lip synthesis network supporting side-face views; eliminated phoneme-decomposition latency via direct parameter regression.
- **Deployment:** Applied to arbitrary-identity video generation in TAL's AI classroom roll-call system.

AMAP (Alibaba Group)

Senior Algorithm Engineer — Machine Learning

Beijing, China
2021.08 – 2022.05

ETA optimization: contrastive Link2Vec representation + Transformer feature fusion; nationwide excellent-rate +1%, poor-rate -10% period-over-period.

ACADEMIC PROJECT

Power Energy Prediction via Deep Learning LSTM

Individual Researcher, Institute of Industrial & Systems Engineering, NEU

Shenyang, China
2017.09 – 2018.09

- **LSTM + Seq2Seq:** Designed load forecasting model for steel enterprises; proposed two missing-data imputation strategies (mean-fill and BP-network completion).
- **Attention mechanism:** Integrated attention into Seq2Seq wind power prediction, demonstrating a 1.3% accuracy improvement over the prior approach on the benchmark dataset.

PUBLICATIONS & PATENTS

- [1] Xinyi He, Jin Lang, Yanyan Zhang, Gengnan Zhao, and **Xiang Zhao**. (2022). Power Load Forecasting Method of Iron and Steel Enterprises based on Improved LSTM. *Metallurgical Industry Automation*, 46(3), 48–56.
- [2] **Xiang Zhao**, Yule Li, Wei Xiang. (2021). A Model Training, Image Processing Method, Device, Equipment and Medium. *CN 112767287 A*.
- [3] Lixin Tang, Jin Lang, and **Xiang Zhao**. (2018). A Deep Learning-based Ultra-short-term Wind Power Prediction Method. *CN 109615146 A*.
- [4] Yang Yang, Jin Lang, Jian Wu, Yanyan Zhang, and **Xiang Zhao**. (2023). A Novel Correlation-optimized Deep Learning Method for Wind Speed Forecast. *Submitted*.

SELECTED OPEN SOURCE PROJECTS

- **Triton-Torch-Custom:** Industrial-grade example for deploying PyTorch custom ops to Triton. (github.com/lesliezhao/triton-torch-custom)
- **HeadSwap:** High-fidelity head-swap model implementation. (github.com/lesliezhao/headswap)
- **Simple-Lora:** Lightweight LoRA training codebase on Stable Diffusion. (github.com/lesliezhao/simple-lora)
- **GFPGAN-1024:** GFPGAN training supporting 1024-resolution output. (github.com/lesliezhao/GFPGAN-1024)
- **DCT-NET.Pytorch:** Complete PyTorch reproduction of DCT-NET high-fidelity style transfer. (github.com/lesliezhao/DCT-NET.Pytorch)

SKILLS, HONORS & ACTIVITIES

- **Technical:** Diffusion Models (FLUX, Wan-S2V, SD), GANs (StyleGAN, HiFiFace, GFPGAN), RL fine-tuning (Flow-GRPO), VL/VLM-based reward design, annotation, and multimodal supervision, model distillation (DMD, RCM), Triton custom operator deployment.
- **Programming:** Python · PyTorch (proficient); C/C++ (competent); SQL, MATLAB.
- **Languages:** Chinese (native), English (proficient).
- **Honors:** Third Prize, Fashion AI Global Challenge (Top 7%, Alibaba Cloud, 2018); Second-class Scholarship (Top 10%, NEU, 2013).
- **Activities:** Director, LearnHub Club, Northeastern University (2018.09 – 2019.06).